Cary Li

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EDUCATION

University of Toronto - St. George Campus Sep. 2026 - Present
B.A.Sc. Engineering Science | Toronto, ON

Dr. G.W. Williams Secondary School Sep. 2022 - May 2026
IB Diploma: 41/45 | Ontario Secondary School Diploma | Aurora, ON

PROJECTS

Kera | Hack the Valley Oct. 2026
Python, PyTorch, scikit-learn

- Trained a three-seed PyTorch MLP ensemble (42-16-8-1) for photo-based hemoglobin estimation, using standardized RGB features, ReLU and dropout.
- Used AdamW, Huber loss and five-fold grouped cross-validation; achieved 15.96 g/L mean absolute error on 50 held-out patient records.

ClearPath
TypeScript, React, Next.js, Three.js

- Developed an interactive room planner that helps people explore creative layouts and use space more efficiently through synchronized 3D and top-down views.
- Combined photo-based room estimates, furniture editing and A* routing with collision and clearance checks to compare practical layout alternatives.

Anchor Jun. 2026 - Present
Python, Flask, SQLite, Electron, JavaScript, Tailwind CSS

- Developed a full-stack Windows app for goal and habit tracking, with OS notifications and an AI assistant that prompts progress check-ins.
- Built a Flask and SQLite backend and packaged the HTML/CSS/JavaScript interface with Electron.

Racing Algorithms | UofT Formula Racing and Blue Sky Solar Racing Sep. 2026
C++ | Driverless perception and race strategy

- Built a cone-based pathfinding algorithm for Formula Racing, optimizing path length and straightness.
- Simulated solar-car energy use to search for efficient racing speeds; used local caching to accelerate the search.

EXPERIENCE

Founder | The Fundamental Why Mar. 2025 - Present

- Created a physics and philosophy podcast with 7 episodes, 2,100+ subscribers and 14,100+ views.
- Interviewed STEM researchers, physics students and a nonprofit founder across YouTube and podcast platforms.

Competitive Swim Instructor | Champion Swim School Sep. 2024 - Jun. 2026

- Taught all four competitive strokes to ages 6-16 in classes of six; designed drills for 9.5 teaching hours per week.
- Named Instructor of the Month among 30 instructors in October 2025 and February 2026.

AWARDS

UofT National Book Award (2026): Sole recipient among 387 graduating students, recognized for academic achievement, creative thinking and breadth of extracurricular involvement.
Pay It Forward Scholarship (2026): Awarded $1,000 as one of the top five community leaders in Aurora and Newmarket by Magna International Inc. and Neighbour Media.
DECA ICDC Qualifier (2024): Led a team of three on Terra Fuel, a biomass-to-energy concept using pyrolysis and carbon filtration to target net-zero emissions; advanced past 300+ competitors to the international stage.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript, SQL, Java, C, C#, HTML/CSS
Frameworks and tools: PyTorch, scikit-learn, React, Next.js, Three.js, Flask, SQLite, Git